Fiscal Year:	FY 2008	**Task Last Updated:**	FY 02/16/2011
PI Name:	Putcha, Lakshmi Ph.D.		
Project Title:	Assessment of Pharmaceutical Stability in Space Using Ground-based Simulation of Spacecraft Environmental Factors		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:	Deceased as of September 2015.		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	05/15/2004	**End Date:**	09/01/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	3
No. of PhD Candidates:	0	**No. of Master' Degrees:**	2
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/1/2008 per HRP Master Task List information dtd 1/11/2012 (Ed., 4/13/12)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Du, Brian (Wyle Laboratories)		
Grant/Contract No.:	not available		
Performance Goal No.:			
Performance Goal Text:			
	Therapeutic efficacy and safety of pharmaceuticals flown on the space shuttle and International Space Station (ISS) remain critical issues for successful NASA medical operations. Preliminary studies indicated significant differences in potency of some pharmaceutical preparations flown on space flights compared to ground-based controls. This project addresses Question number 11 of Clinical Capabilities in the Critical Path Roadmap. We hypothesize that space flight environment, especially radiation, affects stability of pharmaceuticals contained in the space shuttle and ISS medical kits. The objectives are to 1) determine chemical and physical stability parameters of select pharmaceuticals contained in the shuttle and ISS medical kits along with matching ground-based controls; 2) characterize degradation profiles of the unstable formulations after exposure to particulate radiation using synchrotron light and radiation sources; and 3) develop mathematical models that can predict shelf life of pharmaceuticals during		

Task Description:	space flight. Select dosage forms from the medical kits returned to Earth after a space flight will be analyzed and compared to matching formulations stored under controlled conditions on Earth. Disintegration, dissolution rates and concentrations of active drug and degradation products of these dosage forms will be determined. Effects of high-energy radioactive particles on the stability of these pharmaceuticals will be examined by exposing the formulations to synchrotron light and radiation sources at Brookhaven National Laboratories. Predictive degradation models will be constructed from these data to estimate the degree and magnitude of loss of pharmaceutical activity during space flights. In addition to providing information on pharmaceutical stability in space, the proposed research will also identify if there is a need to develop special pharmaceutical preparations that can ensure stability and resultant therapeutic efficacy of pharmaceuticals in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research expands and adds to current understanding of medication and dosage form stability. The primary customer inside NASA is the astronaut flight crew and Flight Medicine by improving and optimizing medical kit configuration. Understanding medication stability in spaceflight benefits the crew by ensuring treatment success. The primary application of this research outside NASA, primarily by optimization of medicine storage or optimization of medical kit configuration in special environments such as military, maritime, and remote and isolated research outposts.
Task Progress:	[Editor's note 2/16/2011: Compiled and submitted for the PI with her permission, from Final Technical Report submission in 2010] The results of this study demonstrate beyond a doubt, that there are certain formulations contained in the current medical kits that depict less than clinically desirable shelf-life characteristics in space and this difference in stability may be due to specific factors in space affecting pharmaceutical stability. Results from NXPCM payload study on stability of Pharmacotherapeutic and Nutritional Compounds will provide information on significant differences in drug stability under controlled conditions on ground and in space. However, in this study, we identified the medications with suboptimal stability characteristics that may compromise therapeutic efficacy. On the other hand, all 3 of the topical preparations from the SOMS and ISS kits, ciprofloxacin solution, desoximetasone ointment and triamcinolone cream, typically flown in their sealed commercial dispensers, appear to have comparable shelf-life and stability variables in flight and on ground. These results suggest that formulations in their original commercial packaging may be more stable in space than those in special flight packaging. These results, however, can be confirmed once analysis of the remaining samples for these formulations is completed. Comparison of results on formulations returned from SOMS and ISS kits with those from the Payload experiment will provide a better understanding of factors that contribute to changes in stability indicating parameters during space missions.
Bibliography Type:	Description: (Last Updated: 06/23/2025)